PNOZmulti TCP/IP communication with SIMATIC PLCs in Step7 V5.5

Product
Type: PNOZmulti with Ethernet-Interface (TCP/IP)
Name: PNOZmulti series (complete list inside, please see Ch. 3.2, page 8)
Manufacturer: Pilz GmbH & Co. KG, Safe Automation

Document
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Document Revision History

Release	Date	Changes	Chapter
01	2012-08-23	Creation	all
02	2014-06-16	Insert PNOZ m ES ETH Module	all

Validity of Application Note

This present Application Note is valid until a new version of the document is published.
This and other Application Notes can be downloaded in the latest version and for free from www.pilz.com.
For a simple search, use our content document (1002400) or the direct search function in the download area.

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We are grateful for any feedback on the contents.

June 2014

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You can reach our international hotline on:

+49 711 3409-444 or **mailto:support@pilz.com**

Pilz GmbH & Co. KG
Safe Automation
Felix-Wankel-Straße 2
73760 Ostfildern, Germany

Telephone: +49 711 3409-0
Telefax: +49 711 3409-133
E-Mail: pilz.gmbh@pilz.de
Internet: www.pilz.com

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1. Useful documentation

Reading the documentation listed below is necessary for understanding this application note. The availability of the indicated tools and safe handling are also presupposed with the user.

1.1. Documentation from Pilz GmbH & Co. KG

No.	Description	Item No.
1	Pilz international homepage, download section	www.pilz.com
2	PNOZmulti Communication Interfaces	1001154-EN-XX
3	PNOZmulti 2 Communication Interfaces	1002971-EN-XX
4	Application Note PNOZm Serial-2-Ethernet	1001958-EN-XX
5	Technical Catalogue PNOZmulti	1001153-EN-XX
6	Operating Manual PNOZ mm0.1p	1002101-EN-XX
7	Operating Manual PNOZ mm0.2p	1002235-EN-XX
8	Operating Manual PNOZ mmc1p	1002102-EN-XX
9	Operating Manual PNOZ m1p (ETH)	20878-EN-XX
10	Operating Manual PNOZ m B0	1002660-EN-XX
11	Operating Manual PNOZ m ES ETH	1002700-EN-XX

1.2. Documentation from other sources of information

No.	Description	Item No.
1	SIMATIC portal “Automation Systems” (international) Internet-Link to "www.automation.siemens.com"	
2	Document, Code and Step7-Project to topic “Wizard for creating connection data for open TCP/IP communication (Open Communication Wizard).” Internet-Link to "support.automation.siemens.com"	OC Wizard V2.3.3 Entrys-ID: 25209116

2. Preface

This document provides a Step by Step example how to communicate between Siemens S7-300 and PNOZ multi with TCP/IP connection.

Notice:
This Application Note describes a PNOZ m1p ETH, the PNOZ mmc1p is working in the same way. With a PNOZ m B0 ETH you have to take different Blocks (because of bigger Data Size) in the Simatic Example, it is also included in the Simatic Example.

The document is not intended as a technical documentation for general use of SIMATIC Manager Step 7. It may not be distributed to customers without a special note to its scope.

Notice: You need for this application these four projects

- PNOZmulti Configurator – provided Project
  - "**PNOZmulti_S7-PLC_ETH-TCP-IP_v2**"
  - "**PNOZ_mmc1p_S7-PLC_ETH-TCP-IP_v1**"
  - "**PNOZ_m_ES_ETH_S7-PLC_ETH-TCP-IP_v1**"
- Step7 – provided Project
  - "**PNOZm_S7-PLC_ETH_v2**"

This following Step7-Project is not provided. It is an example for your second target project.

- Step7 – Target Project Example "ETH_Multi_S7_ProfiNET_v2"

In addition to this manual the first **four projects**, which are described in the following, can be opened with a unzip program from the attached file

"**AN_PNOZm_ETH_TCP-IP_with_S7_1002671_SW_02.zip**"

by a double-click on this "Pin-Icon": » «

Notice:
It is not possible to open or save a *.zip File from the attachments with Adobe Reader.
>> For open or save this file, please use PDF-XChange Viewer (www.tracker-software.com).

Alternatively the file can be obtained from Pilz Technical Support (mailto:support@pilz.de)
Subject: Application Note 1002671

3. Configuration

3.1. List of used hardware

This Application Note has been tested with the following hardware:

- Siemens Simatic PLC: CPU 318-3FL01-0AB0 V3.2
- Pilz Configurable Control System: PNOZ m1p ETH 773104 Version v2.1
- Pilz Configurable Control System: PNOZ m B0 772100 Version v1.1
- Pilz Configurable Control System: PNOZ m ES ETH 772103 Version v1.0

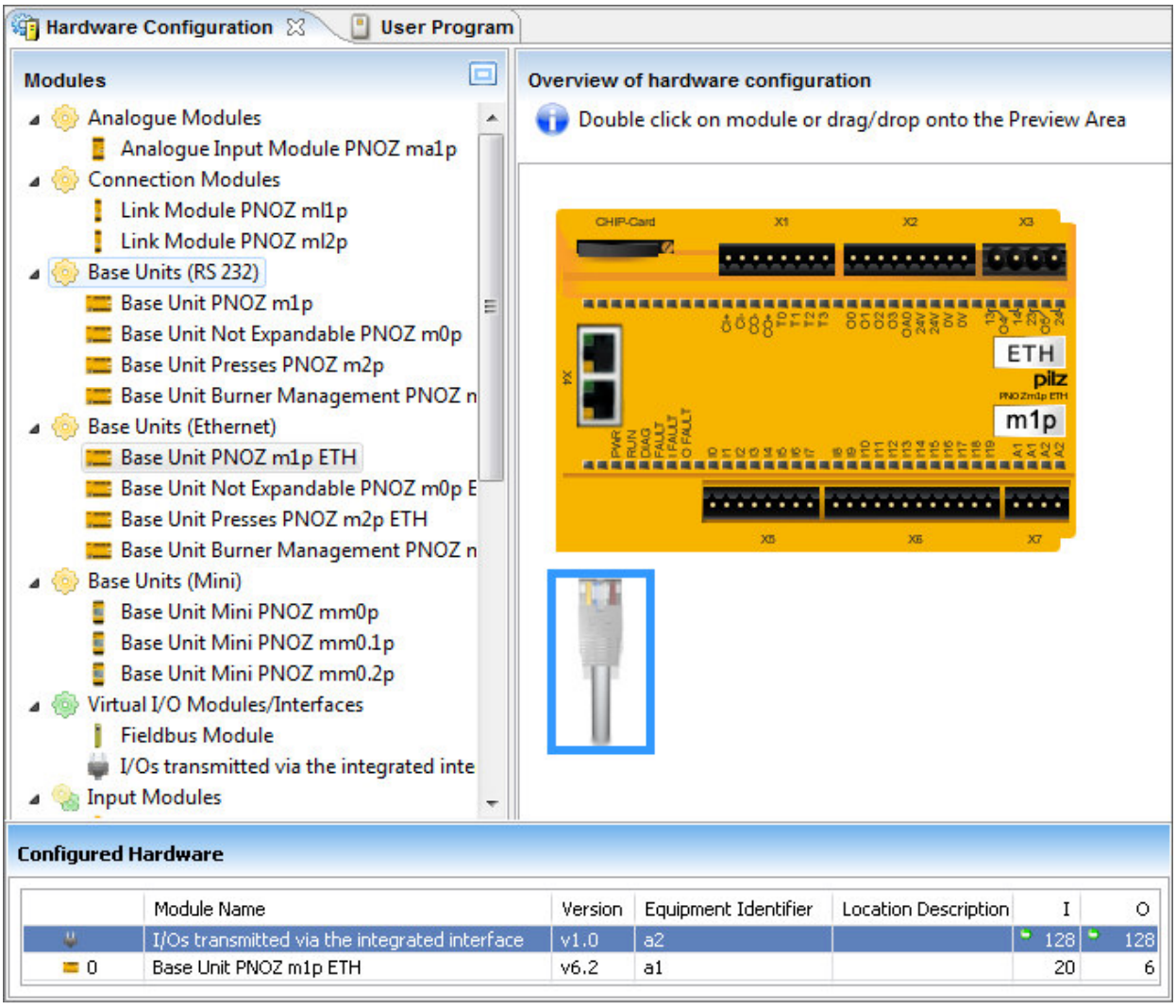


Fig. 1: PNOZmulti Configurator - Hardware Configuration PNOZ m1p ETH

3.2. List of supported Pilz hardware

- PNOZmulti Mini Series
  - PNOZ mm0.1p 772001 with PNOZ mmc1p ETH 772030
  - PNOZ mm0.2p 772002 with PNOZ mmc1p ETH 772030

- PNOZmulti 1st Series with integrated Ethernet ports
  - PNOZ m1p ETH 773103
  - PNOZ m3p ETH 773126
  - PNOZ m0p ETH 773113
  - PNOZ m2p ETH 773123
  - PNOZ m1p ETH coated version 773104

- PNOZmulti 1st Series with external Ethernet ports

Notice: The description of different structure of data telegram of the following configurations is not part of this document. For more information please see Ch. 1.1-No. 2, page 5.

  - PNOZ m1p base unit 773100
  - PNOZ m1p base unit coated version 773105
  - PNOZ m2p base unit press function 773120
  - PNOZ m3p base unit burner function 773125

- PNOZmulti 2nd Series B0 with (external) Ethernet ports
  - PNOZ m B0 772100
  - PNOZ m ES ETH 772130

3.3. List of used Software

This Application Note has been tested with the following software:

- Pilz PNOZmulti Configurator, Version 9.2.0
- Siemens Simatic STEP7, Version 5.5 SP3
- Siemens Simatic Open Communication Wizard, Version 2.3

3.4. List of used IP addresses

The following IP addresses has been used in the example project:

- Computer: 169.254.060.099
- PNOZmulti ETH: 169.254.060.001
- PNOZ m ES ETH: 169.254.060.002
- Simatic S7 PLC: 169.254.060.151

Notice: For more information about setting IP address open in software “PNOZmulti Configurator” Help function "Contents and Index" and choose "Create or edit project".

3.5. List of used S7 and PNOZ multi Programs according to PNOZ multi Hardware

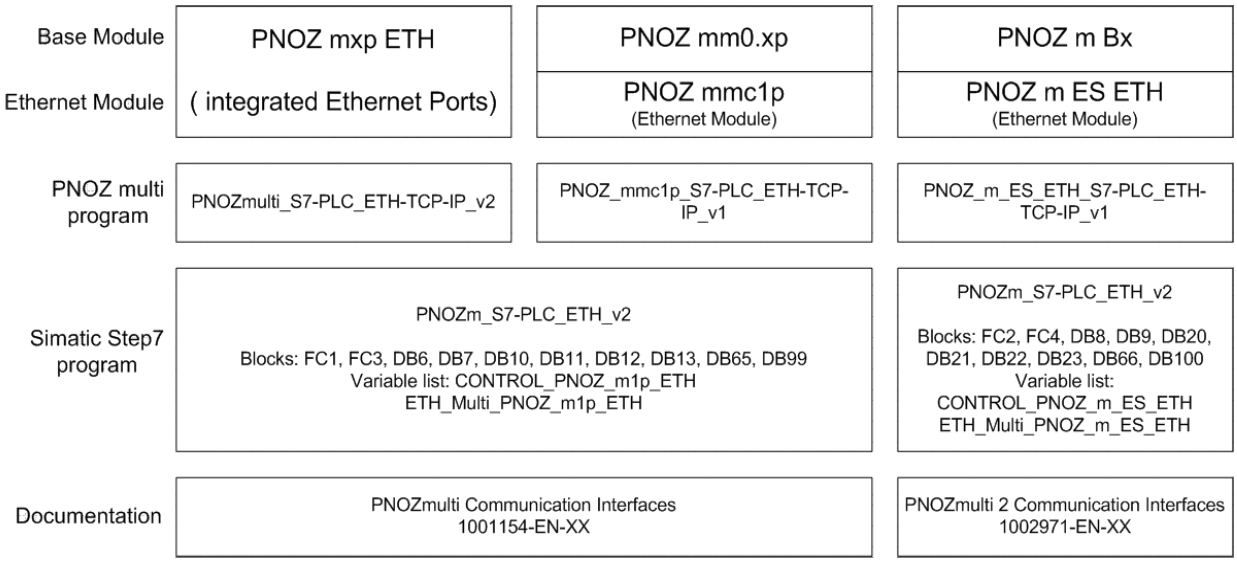


Fig. 2: List of used Programs

4. Application Task

Create a connection with TCP/IP and control communication between PNOZ m1p ETH / PNOZ m ES ETH and Siemens S7-300 PLC.
The initial S7-Hardware configuration is not part of this “Application Note”. One can get further details in the manuals about Siemens Simatic Step 7 (please see Ch. 1.2, page 5).

4.1. Setting IP address on PNOZmulti base unit

- Start the PNOZmulti Configurator and open “Scan Network”.

Fig. 3: PNOZmulti Configurator - Scan Network

- Choose your device and go online. For that click on the “Online”-button.

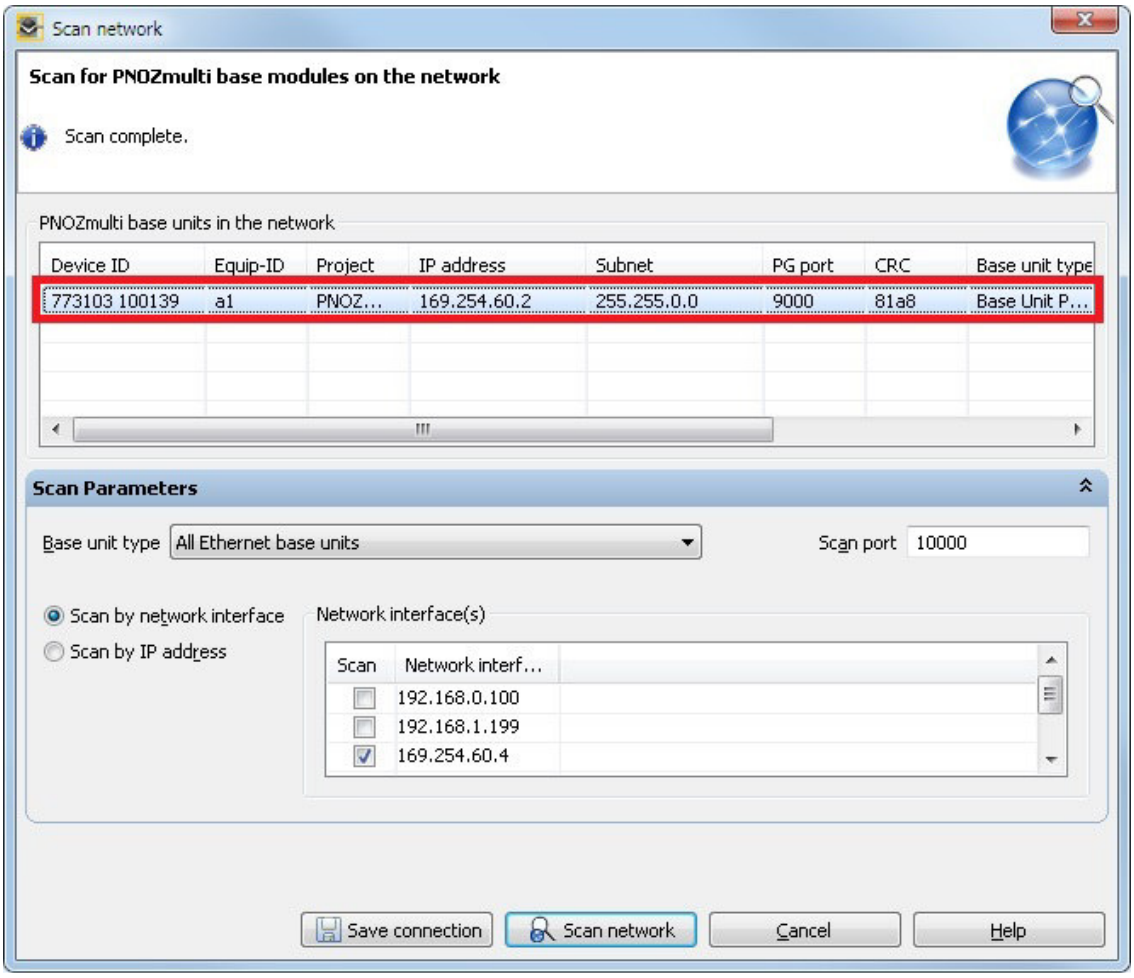


Fig. 4: PNOZmulti Configurator - Choose of the used device

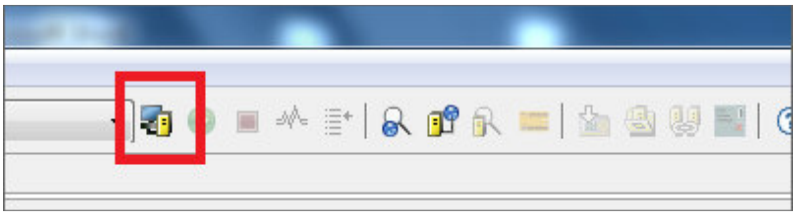


Fig. 5: PNOZmulti Configurator - Go online with the device

If necessary adjust the IP address of your PNOZmulti.

- Continue with function "Configure Device Ethernet Connection".

Fig. 6: PNOZmulti Configurator - Configure device Ethernet connection

- Change the IP address of your device und confirm with "OK".

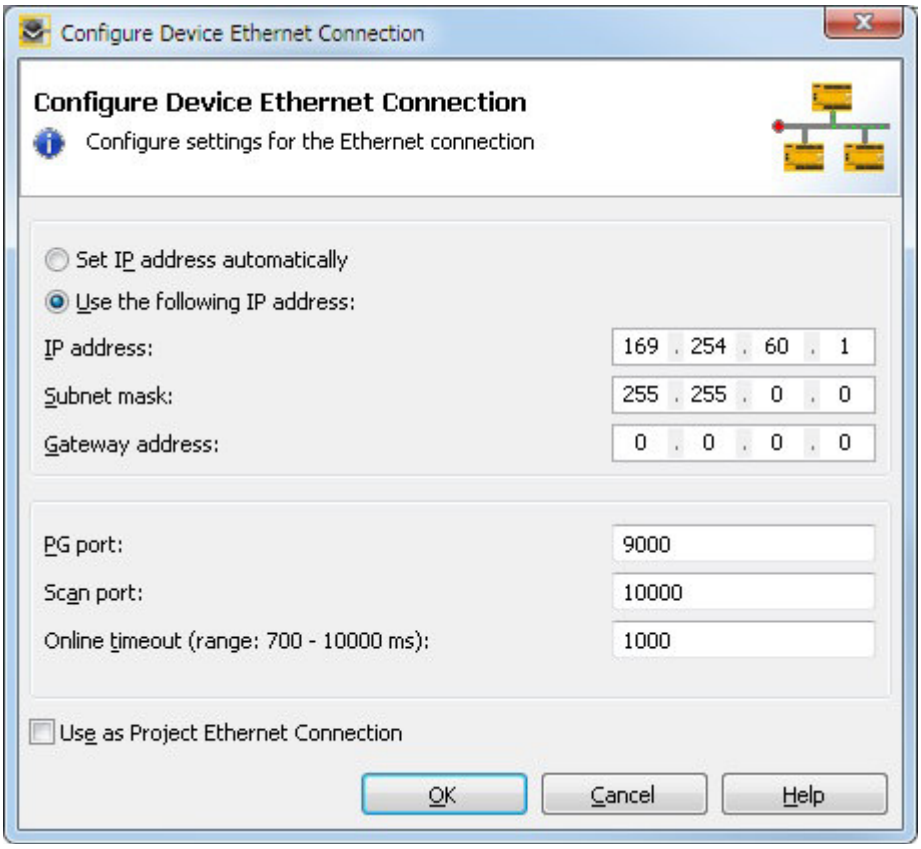


Fig. 7: PNOZmulti Configurator - Adapt the IP address on base unit

The defined IP address is now set to PNOZmulti. The device wills response to that IP address.

4.2. Setting IP address on Simatic PLC (ProfiNET)

- Open the provided Step7-Project “PNOZmulti_S7-PLC_ETH_v2” in the Simatic Manager.
- Copy the blocks this 1st project to your actual project (here as example “ETH_Multi_S7_ProfiNET_v2”) and click twice at “Hardware”.
- Click twice on PN-IO (X2). The window "Properties - PN-IO" open.

If necessary improve the device name to "PN-IO" **[1]**.

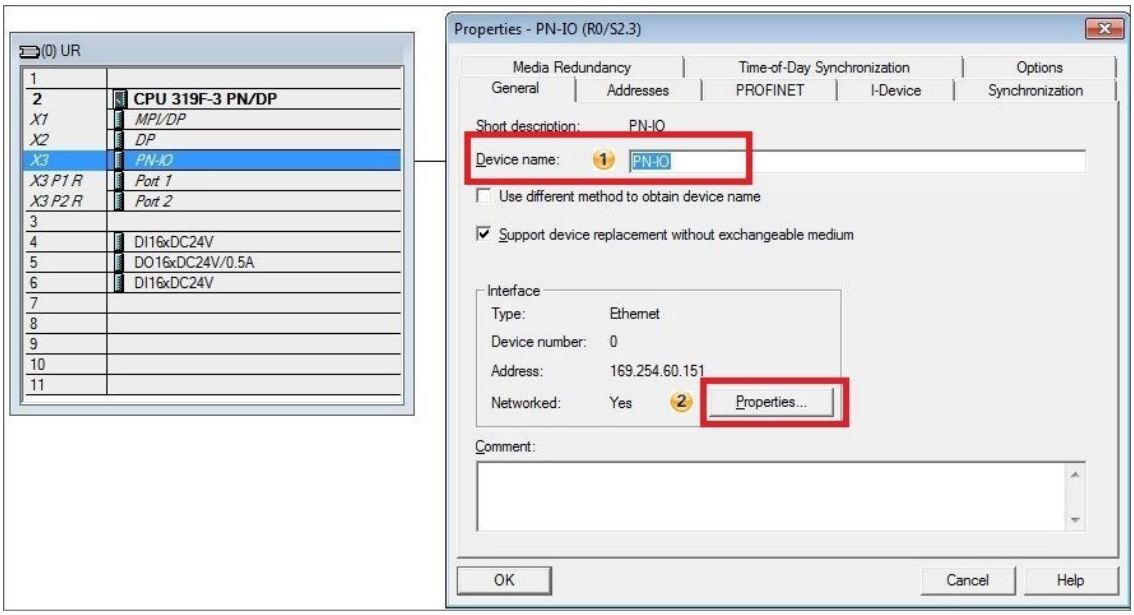


Fig. 8: Simatic HW Config - Setting device name

- Click on the Button "Properties…" if you have to change the IP address **[2]**.

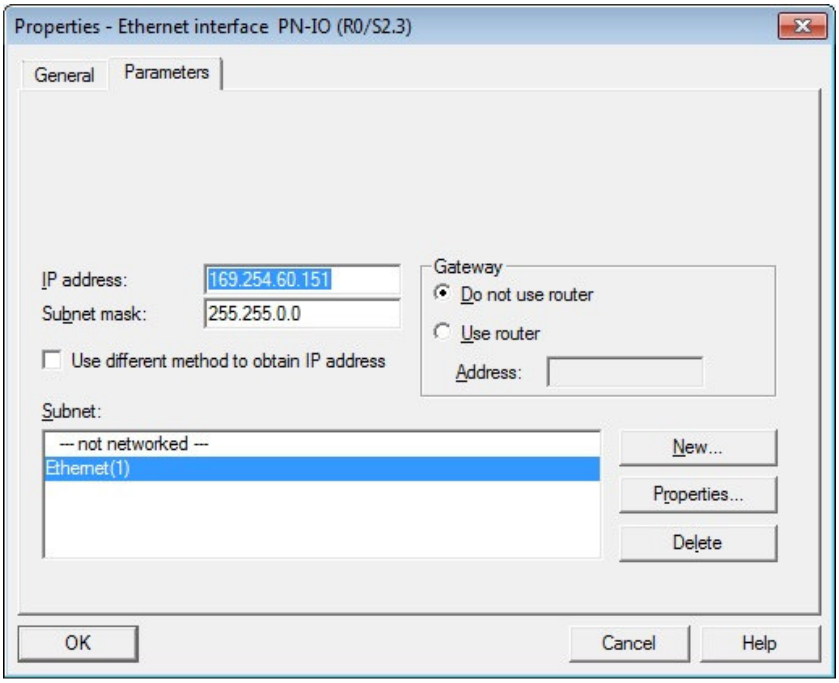


Fig. 9: Simatic HW Config - Change the IP-address

4.3. Settings with Open Communication Wizard

- Start the "Open Communication Wizard" and click at "Next".

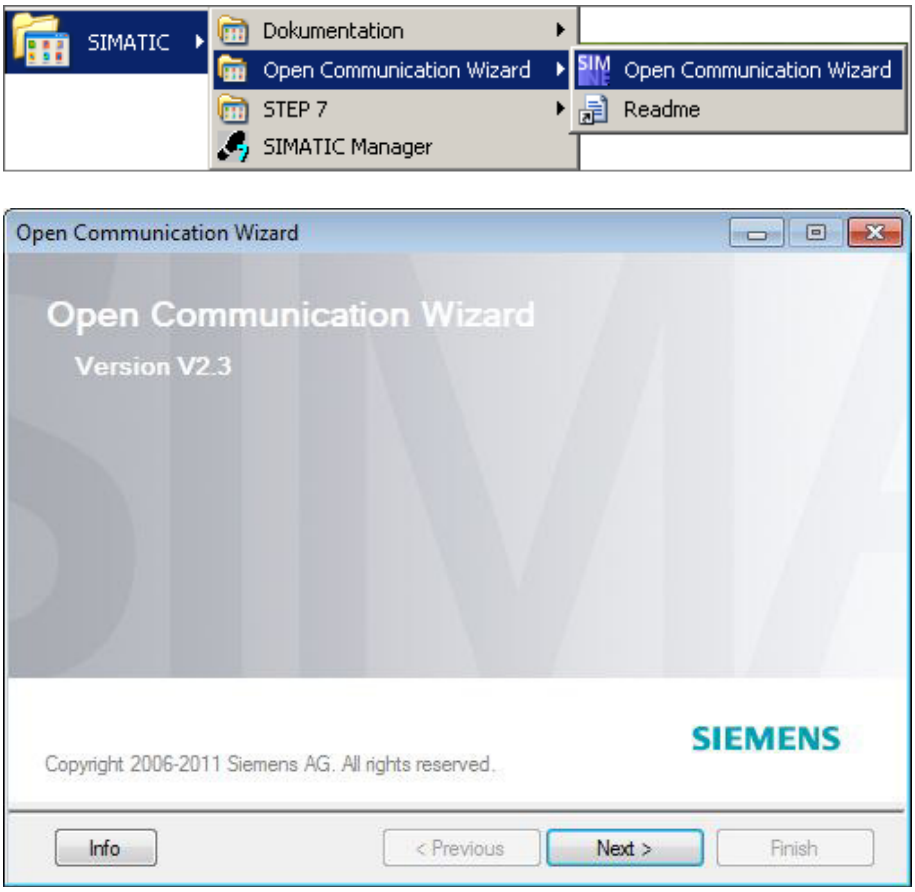


Fig. 10: OC Wizard - Start Open Communication Wizard

- Choose your S7 project and your block folder and confirm with "Open".
- Click "Next" to continue.

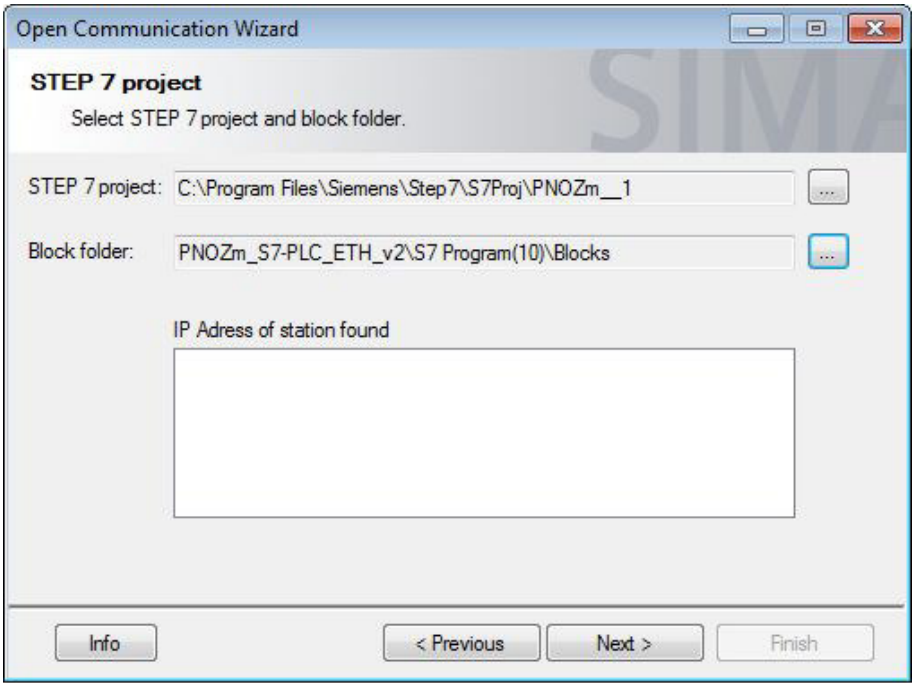


Fig. 11: OC Wizard - Choose the Step 7 project and block folder

- Click "Next" to continue.

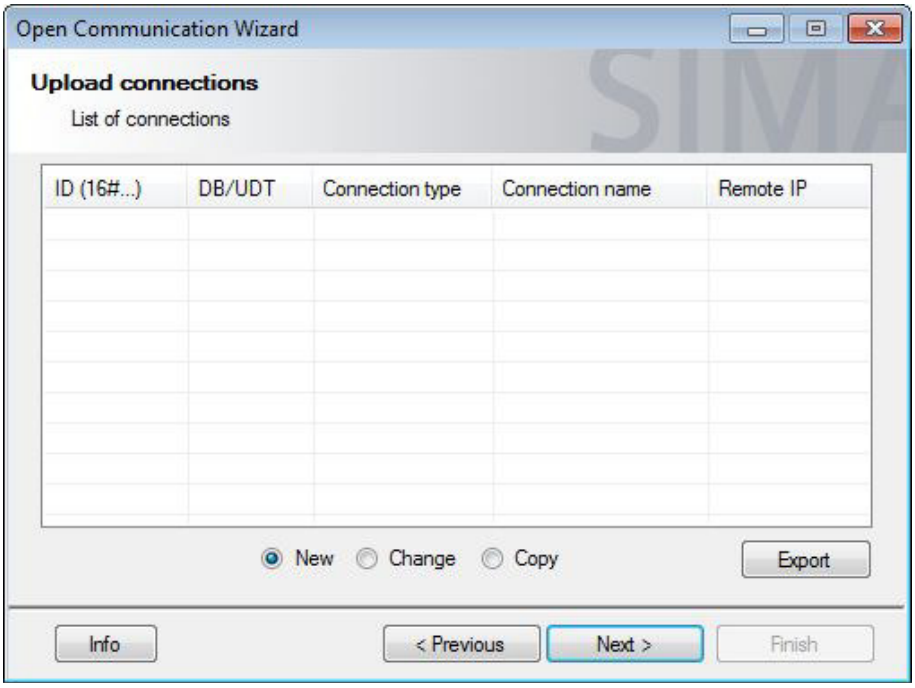


Fig. 12: OC Wizard – Check connection

- Choose "TCP native" and go on with "Next".

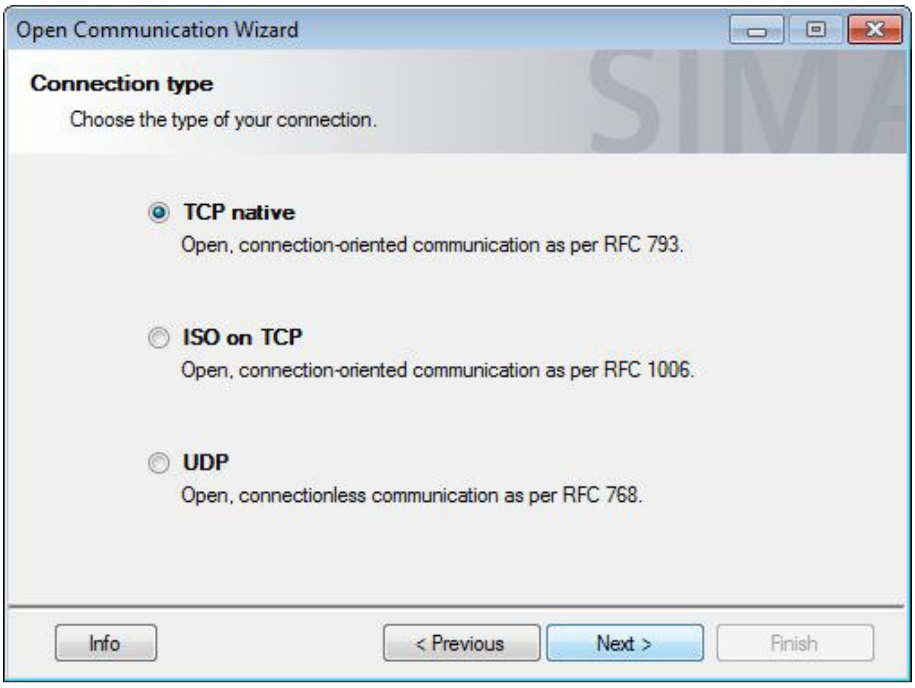


Fig. 13: OC Wizard – Choose connection type “TCP native”

- The “Communication partner B (here: PNOZmulti base unit) is not a S7-CPU”, so activate the right check box and continue with "Next".

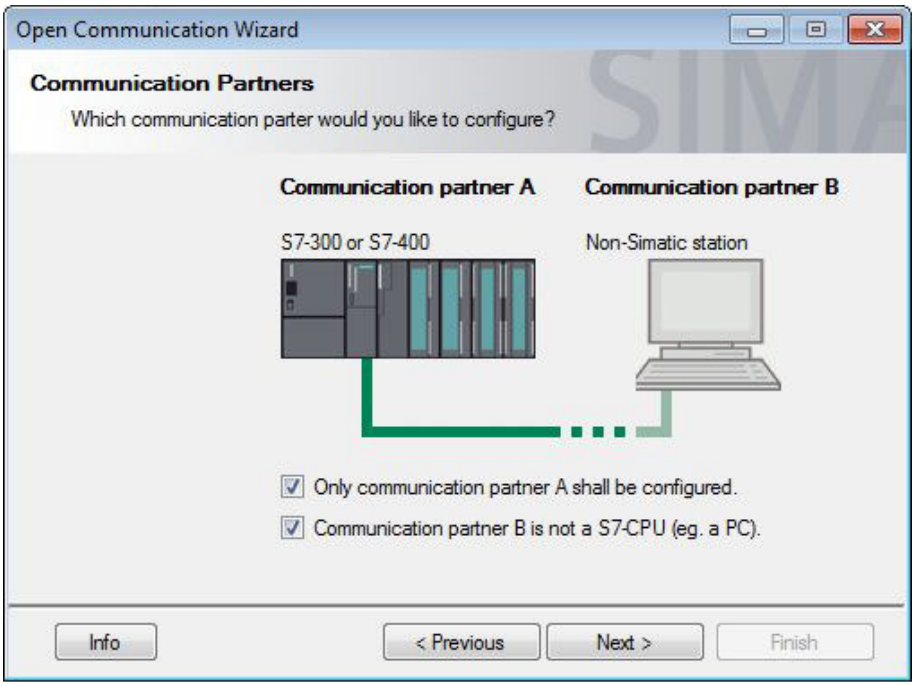


Fig. 14: OC Wizard – Set properties of “Communication partner B”

- Assign the connection name and the IP address and choose the "Used interface" and continue with "Next".

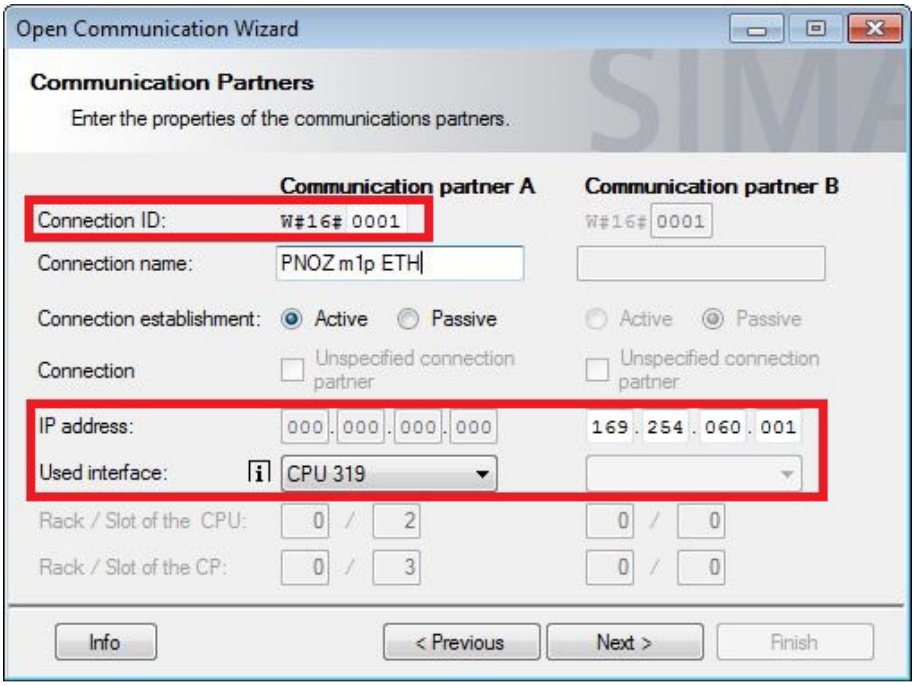


Fig. 15: OC Wizard – Set properties of "Communication partner A"

Normally the port of the PNOZ multi Ethernet modul is the port "9000".

- If the port is changed, please correct this parameter and go on with "Next".

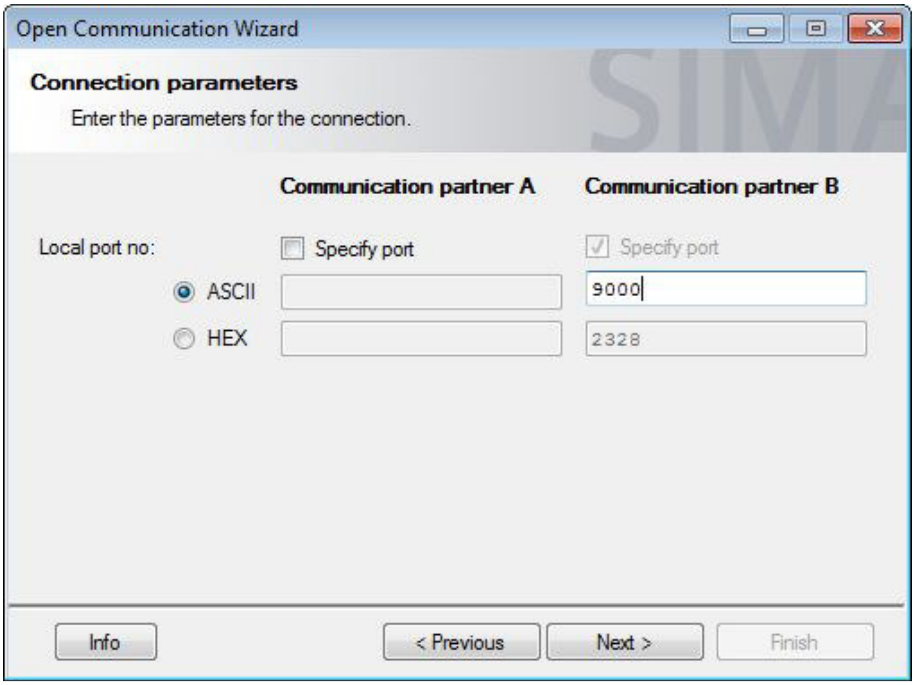


Fig. 16: OC Wizard – Set "Connections parameters" of "Communication partner B"

- Declare any UDT. In this example "UDT 65" and continue with “Next”.

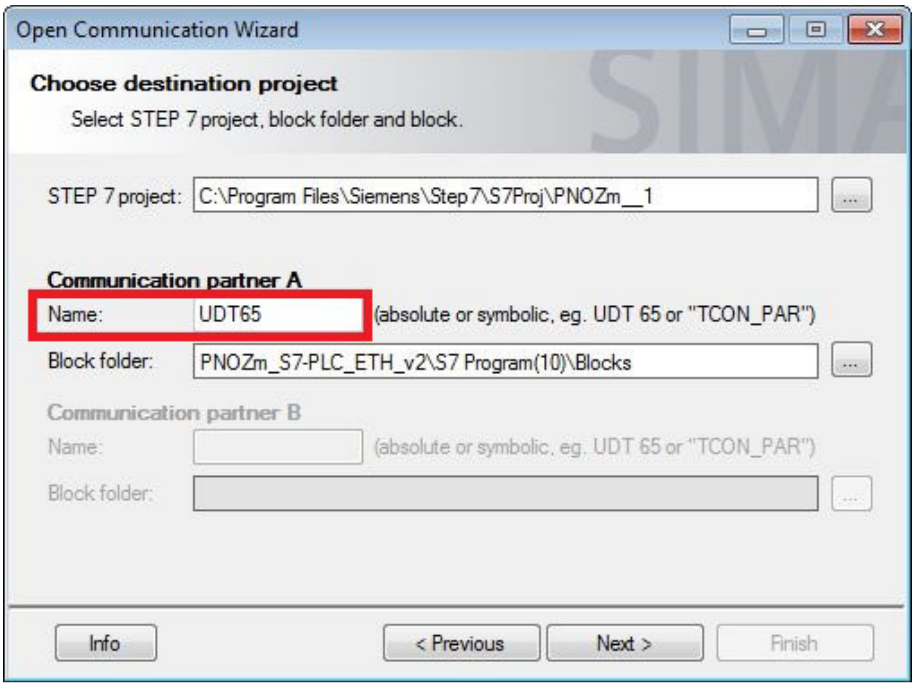


Fig. 17: OC Wizard – Choose destination project, block folder and block “UDT65”

- Control all properties of new connection and continue with "Next".

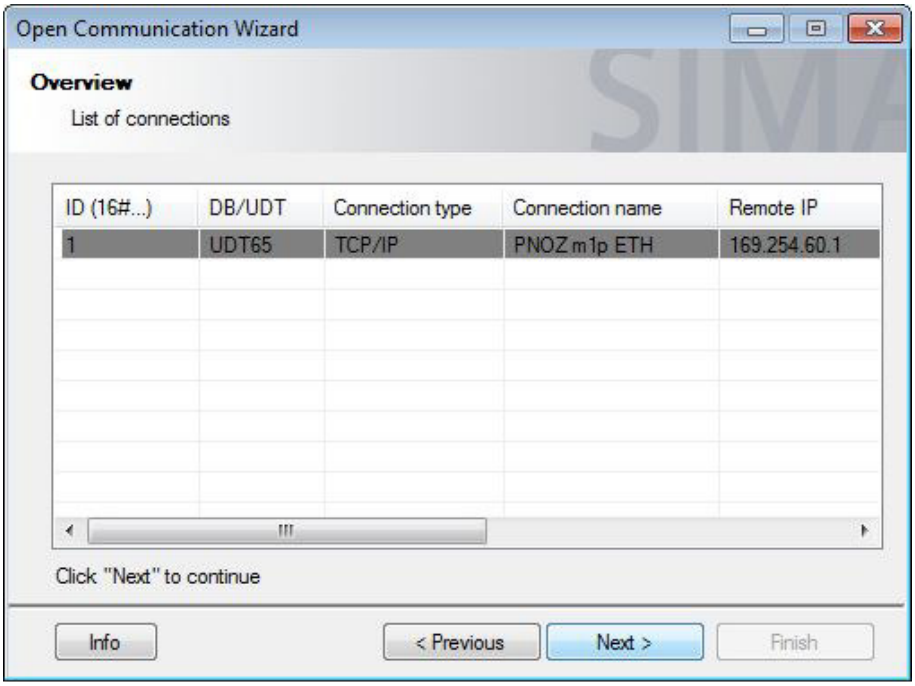
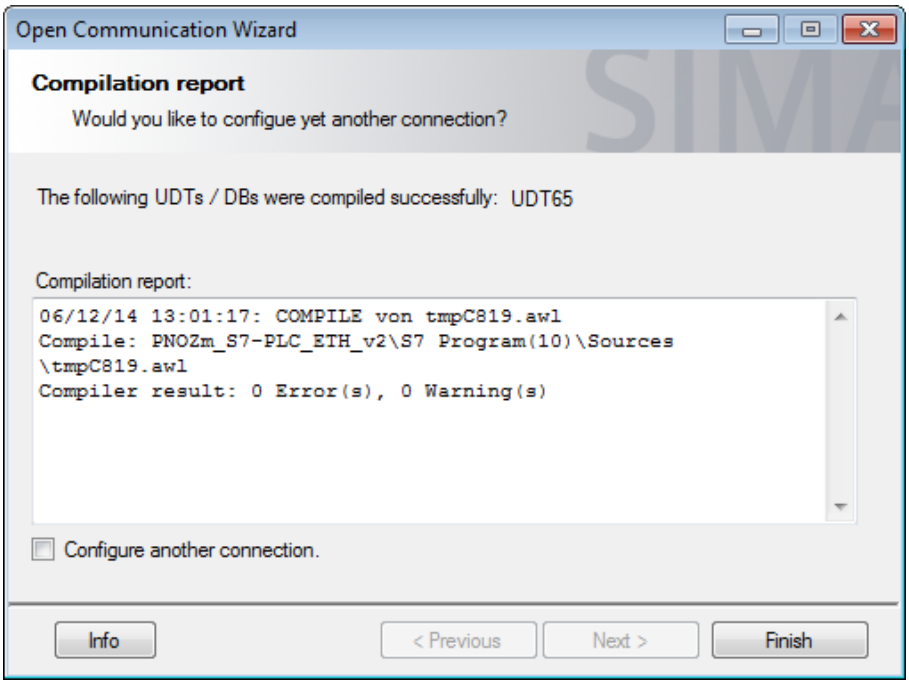


Fig. 18: OC Wizard – Overview with “List of Connections” for “UDT65”

In the S7 project will be set “UDT 65”:

Fig. 19: Simatic Manager – Step7-Project with List of blocks (“UDT65”)

- Confirm with “Finish” and close the “Open Communication Wizard”.



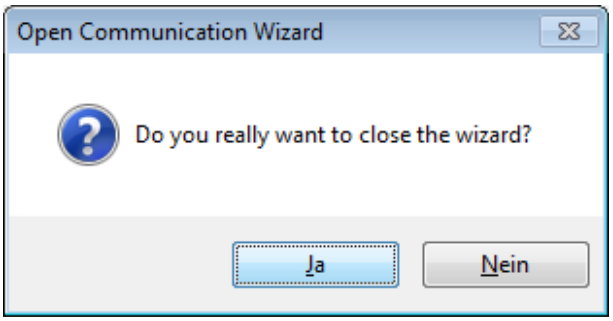
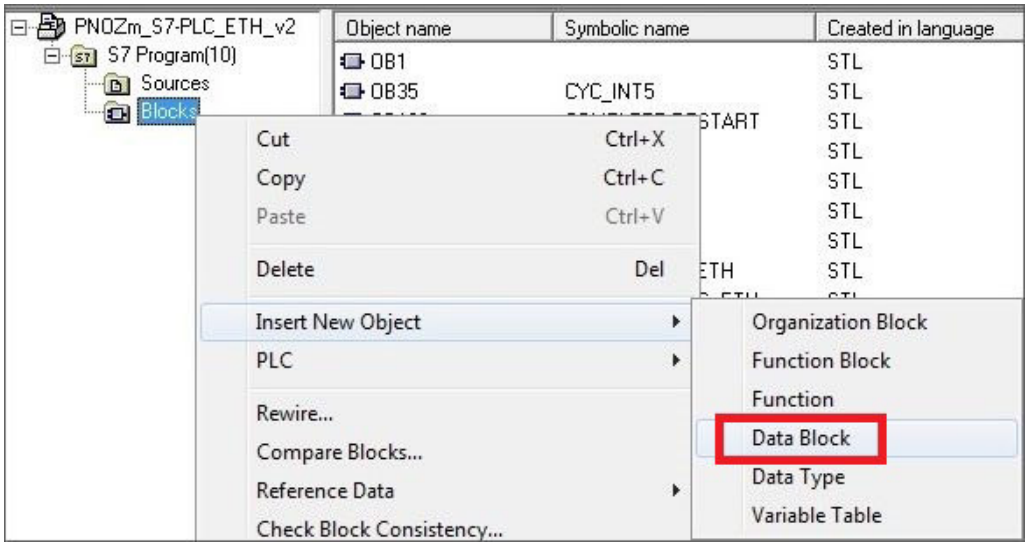


Fig. 20: OC Wizard – Compilation Report and Confirm finished of setup

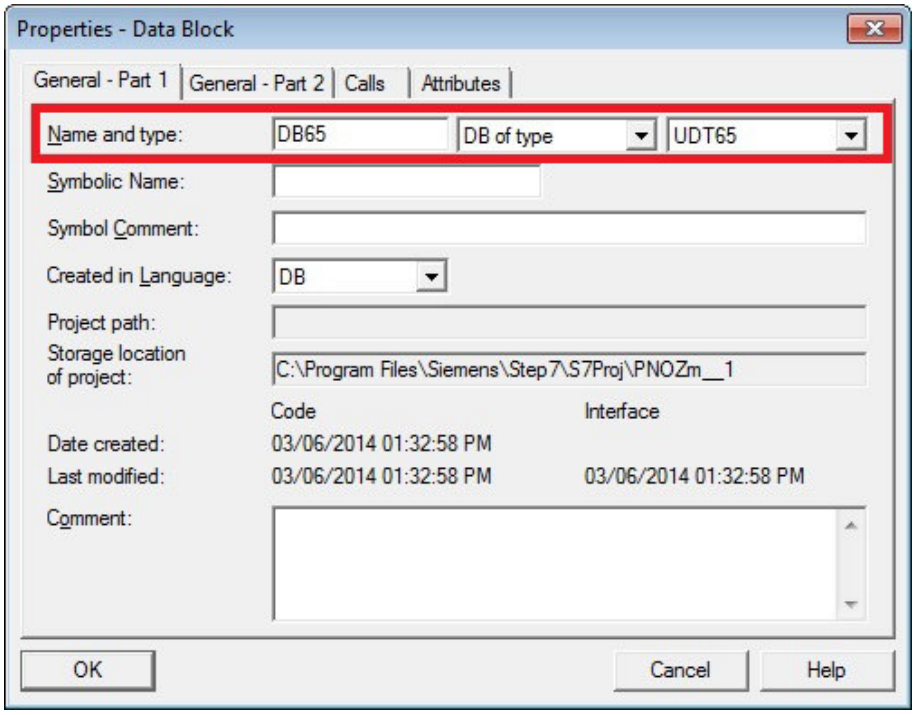
4.4. Complete and Download actual Step7-Project

- Open your actual Step7-Project (here as example “ETH_Multi_S7_ProfiNET_v2”) in the Simatic Manager.
- Now you have to insert a new data block.
  - For this step click with the right mouse button at “Blocks” (German: “Bausteine”), choose "Insert New Object" and click at "Data Block".
  - For properties set name “DB65”, "DB of type" of the data block and “UDT 65”.

Step 1:



Step 2:



Step 3:

Fig. 21: Simatic Manager – Create and set properties of Data Block “DB65” of “UDT65”

- Open Function Block "FC3" for PNOZ m1p ETH, PNOZ mmc1p / “FC4” for PNOZ m ES ETH and change the ID in all network parts.

```
// ==== CONNECT ====================================

      CALL  "TCON" , "InstanzDB-FB65"     FB65 / DB10        -- Connect
       REQ     :=M10.0
       ID      :=W#16#1
       DONE    :=M10.1
       BUSY    :=M10.2
       ERROR   :=M10.3
       STATUS  :=MW11
       CONNECT:=P#DB65.DBX0.0 BYTE 64
```

Fig. 22: AWL-Editor – Change ID in Function Block “FC3”, section “CONNECT”

```
// ==== SEND ===================================

      CALL  "TSEND" , "InstanzDB-FB63"          FB63 / DB11        -- Send Data
       REQ    :=M20.0
       ID     :=W#16#1
       LEN    :=30
       DONE   :=M20.1
       BUSY   :=M20.2
       ERROR  :=M20.3
       STATUS:=MW21
       DATA   :="PNOZ m1p ETH SEND-DATA".DB_VAR    P#DB6.DBX0.0
```

Fig. 23: AWL-Editor – Change ID in Function Block “FC3”, section “SEND”

```
// ==== RECEIVE ================================

      CALL  "TRCV" , "InstanzDB-FB64"           FB64 / DB13        -- Receive Data
       EN_R    :=M40.0
       ID      :=W#16#1
       LEN     :=30
       NDR     :=M40.1
       BUSY    :=M40.2
       ERROR   :=M40.3
       STATUS  :=MW41
       RCVD_LEN:=MW43
       DATA    :="PNOZ m1p ETH RCV-DATA".DB_VAR    P#DB7.DBX0.0
```

Fig. 24: AWL-Editor – Change ID in Function Block “FC3”, section “RECEIVE”

```
// ==== DISCONNECT =============================

      CALL  "TDISCON" , "InstanzDB-FB66"    FB66 / DB12        -- Disconnect
       REQ    :=M30.0
       ID     :=W#16#1
       DONE   :=M30.1
       BUSY   :=M30.2
       ERROR  :=M30.3
       STATUS:=MW31
```

Fig. 25: AWL-Editor – Change ID in Function Block “FC3”, section “DISCONNECT”

- Save modified Function Block "FC3" and download complete Step7-Project to Simatic PLC:

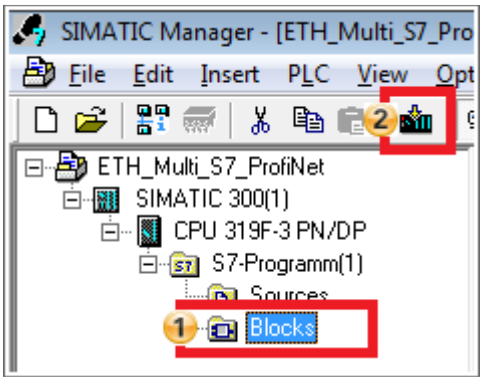


Fig. 26: Simatic Manager – Start complete download of Step7-Project

- For next message box choose “All”:

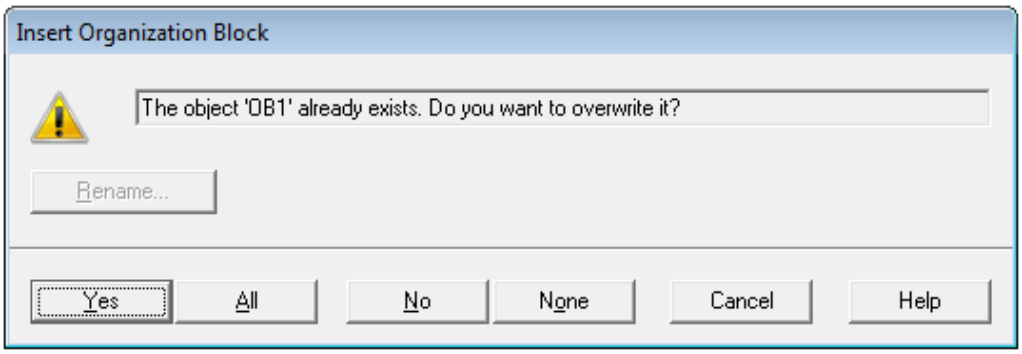


Fig. 27: Simatic Manager – Download, Message box “Insert Organization Block”

- For next four message boxes choose “Yes” or “OK”:

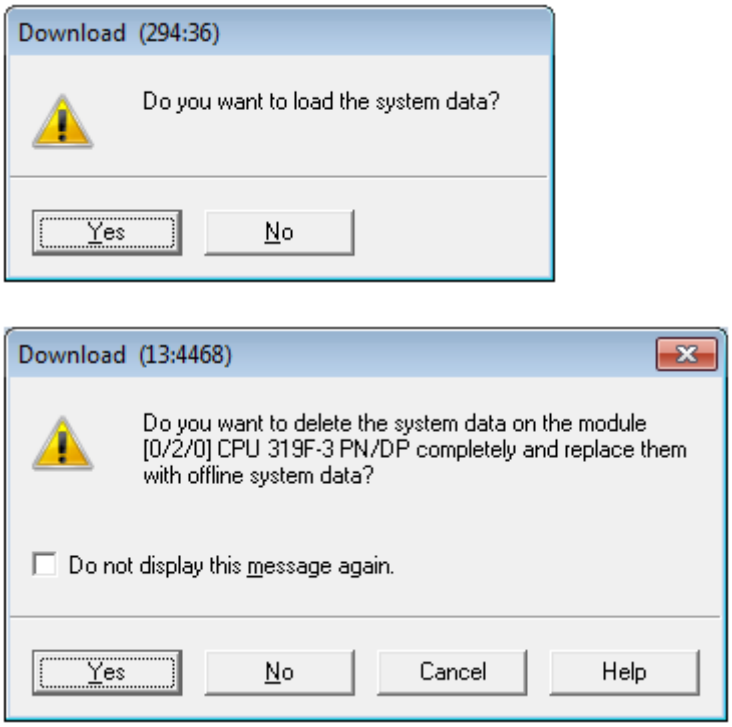


Fig. 28: Simatic Manager – Download, Message boxes “Download”

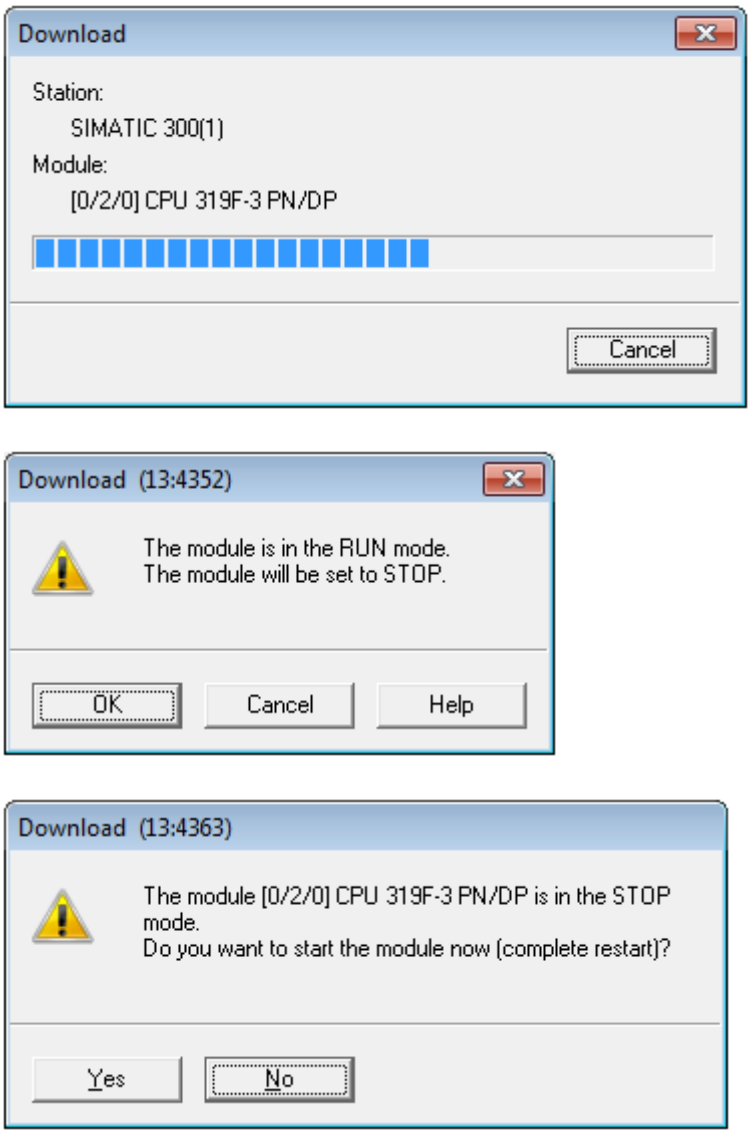


Fig. 29: Simatic Manager – Download, Message boxes "Download"

4.5. Check successful communication via Monitoring

4.5.1. Monitoring on Simatic PLC

- Open both variable tables “CONTROL_PNOZ_m1p_ETH” and “ETH_Multi_PNOZ_m1p_ETH” and Activate for each variable table function “Monitor variable”.

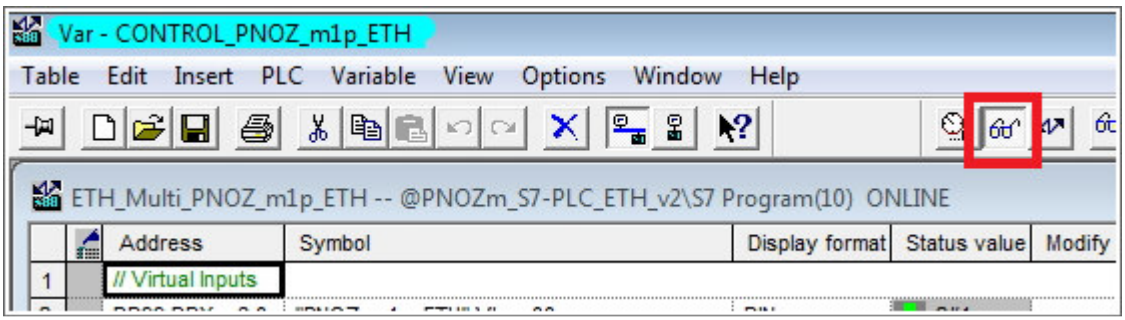


Fig. 30: Var Table – Activate monitoring of variables

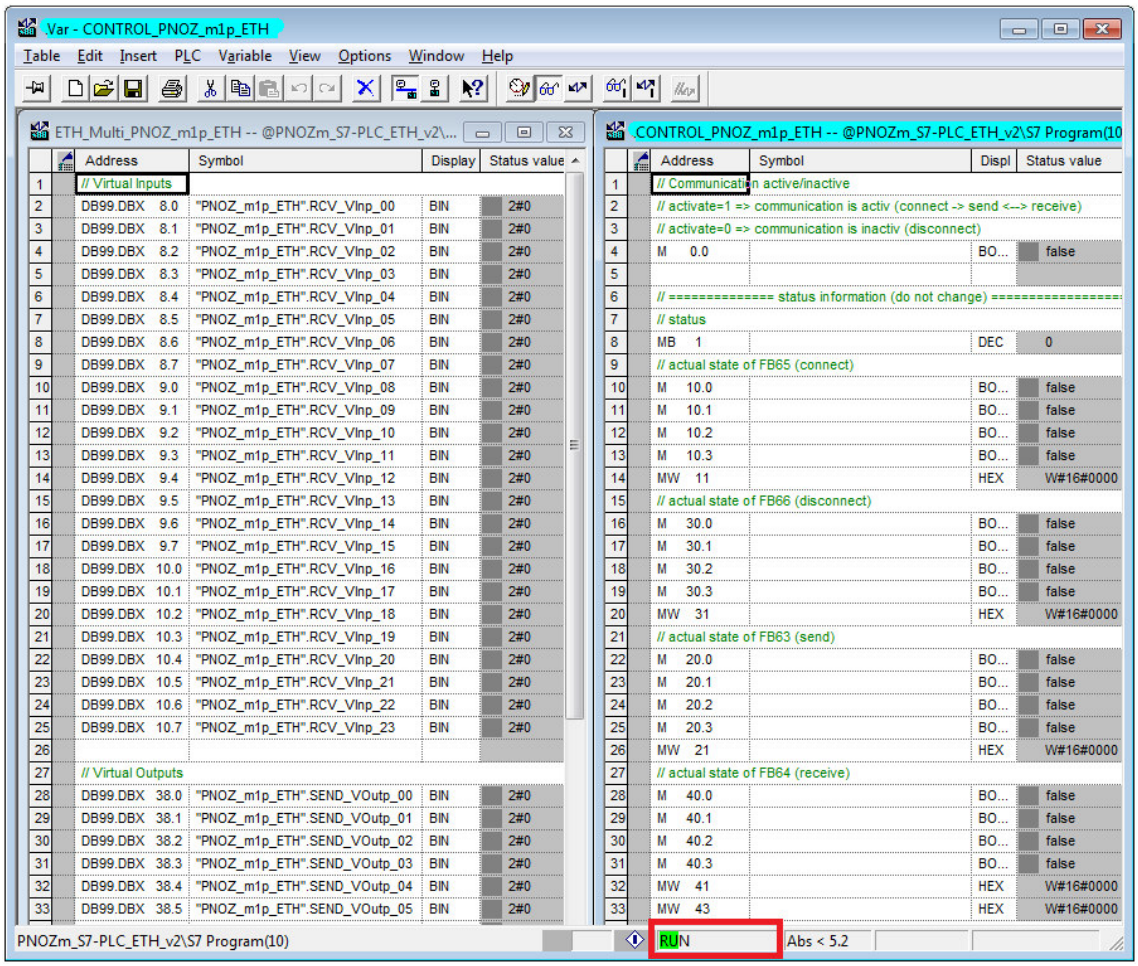


Fig. 31: Var Table – Monitoring of variables is active

- Activate the “CONTROL_PNOZ_m1p_ETH” window, click with your right mouse button at address “M0.0” on Symbol “activate” and set it to ”Modify Address to 1”.

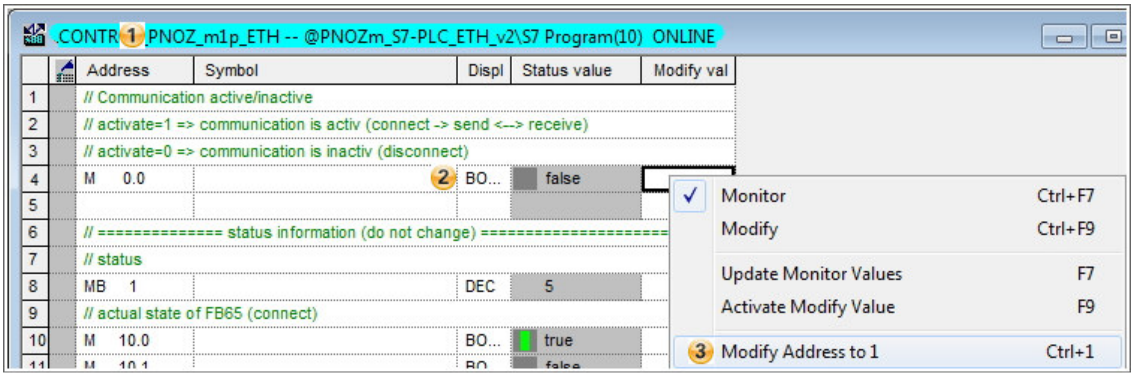


Fig. 32: Var Table – Modify Address “M0.0” to =1

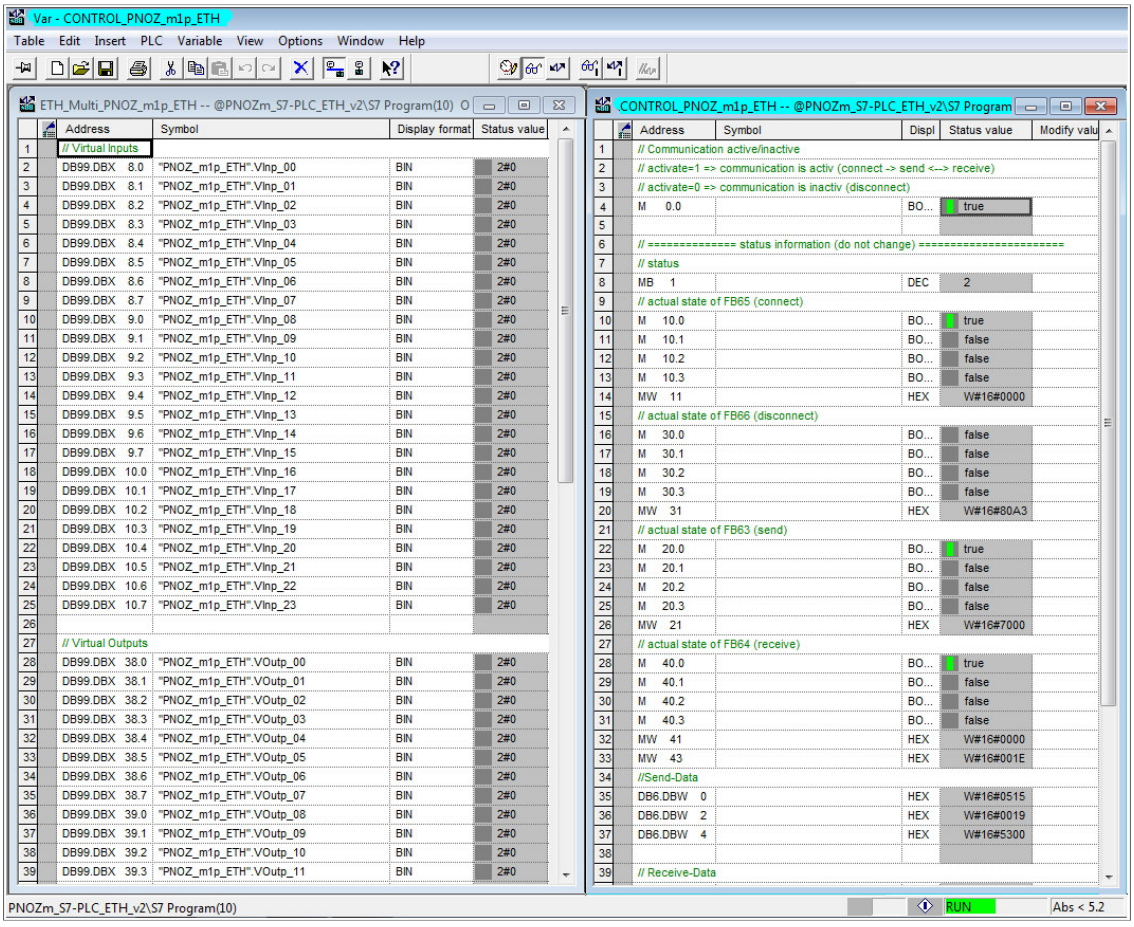


Fig. 33: Var Table – Monitoring of variables is active

- Activate the “ETH_Multi_PNOZ_m1p_ETH” window, select the Virtual Output 00, 08 and 16 and write in the “Modify value” 2#1.

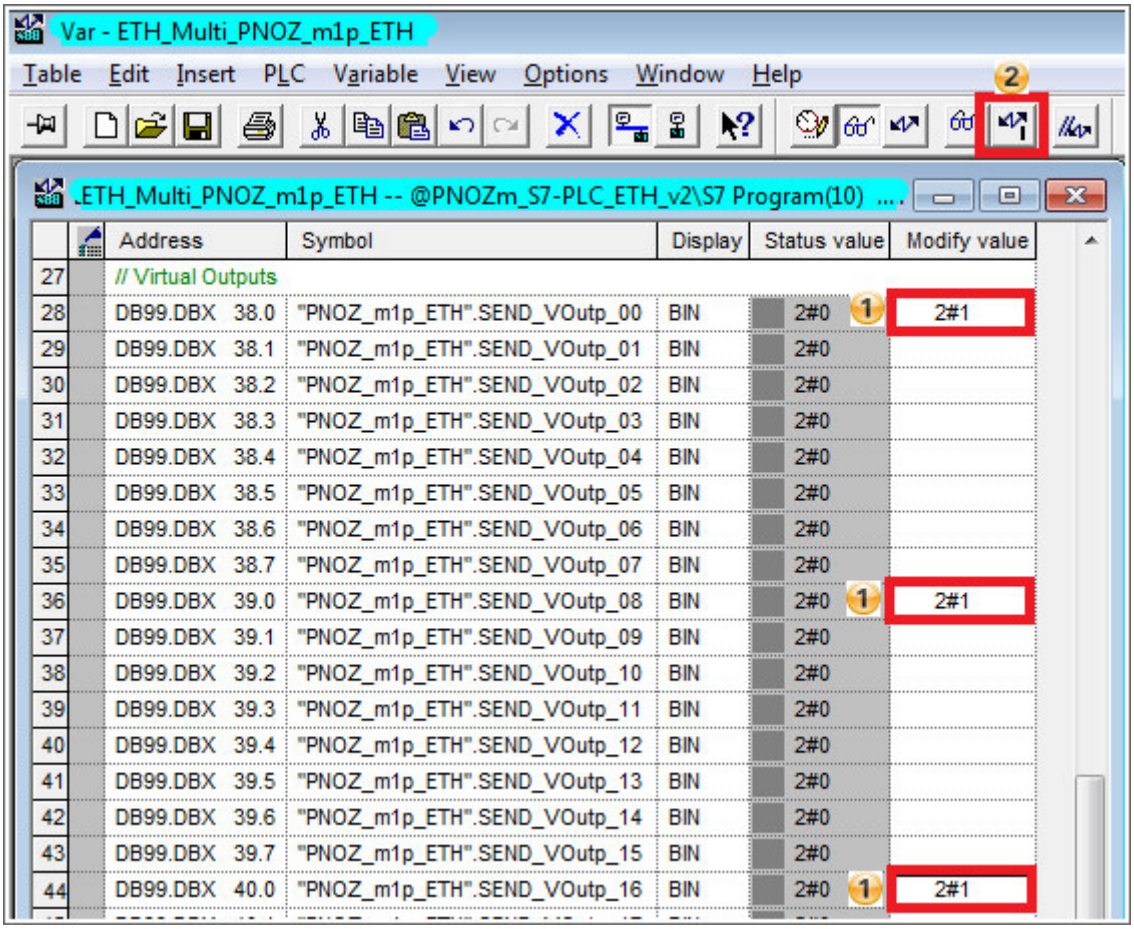


Fig. 34: Var Table – Modify Address Virtual Outputs to =1

The according bits will be set to the signal “=1”.
These tree bits are mirrored in the PNOZ multi application program. When a Virtual Output bit is set to “=1”, the Virtual Input bit is also “=1”.

27		// Virtual Outputs				
28		DB99.DBX 38.0	"PNOZ_m1p_ETH".SEND_VOutp_00	BIN	2#1	2#1
29		DB99.DBX 38.1	"PNOZ_m1p_ETH".SEND_VOutp_01	BIN	2#0	
30		DB99.DBX 38.2	"PNOZ_m1p_ETH".SEND_VOutp_02	BIN	2#0	
31		DB99.DBX 38.3	"PNOZ_m1p_ETH".SEND_VOutp_03	BIN	2#0	
32		DB99.DBX 38.4	"PNOZ_m1p_ETH".SEND_VOutp_04	BIN	2#0	
33		DB99.DBX 38.5	"PNOZ_m1p_ETH".SEND_VOutp_05	BIN	2#0	
34		DB99.DBX 38.6	"PNOZ_m1p_ETH".SEND_VOutp_06	BIN	2#0	
35		DB99.DBX 38.7	"PNOZ_m1p_ETH".SEND_VOutp_07	BIN	2#0	
36		DB99.DBX 39.0	"PNOZ_m1p_ETH".SEND_VOutp_08	BIN	2#1	2#1
37		DB99.DBX 39.1	"PNOZ_m1p_ETH".SEND_VOutp_09	BIN	2#0	
38		DB99.DBX 39.2	"PNOZ_m1p_ETH".SEND_VOutp_10	BIN	2#0	
39		DB99.DBX 39.3	"PNOZ_m1p_ETH".SEND_VOutp_11	BIN	2#0	
40		DB99.DBX 39.4	"PNOZ_m1p_ETH".SEND_VOutp_12	BIN	2#0	
41		DB99.DBX 39.5	"PNOZ_m1p_ETH".SEND_VOutp_13	BIN	2#0	
42		DB99.DBX 39.6	"PNOZ_m1p_ETH".SEND_VOutp_14	BIN	2#0	
43		DB99.DBX 39.7	"PNOZ_m1p_ETH".SEND_VOutp_15	BIN	2#0	
44		DB99.DBX 40.0	"PNOZ_m1p_ETH".SEND_VOutp_16	BIN	2#1	2#1
1		// Virtual Inputs				
2		DB99.DBX 8.0	"PNOZ_m1p_ETH".RCV_VInp_00	BIN	2#1	
3		DB99.DBX 8.1	"PNOZ_m1p_ETH".RCV_VInp_01	BIN	2#0	
4		DB99.DBX 8.2	"PNOZ_m1p_ETH".RCV_VInp_02	BIN	2#0	
5		DB99.DBX 8.3	"PNOZ_m1p_ETH".RCV_VInp_03	BIN	2#0	
6		DB99.DBX 8.4	"PNOZ_m1p_ETH".RCV_VInp_04	BIN	2#0	
7		DB99.DBX 8.5	"PNOZ_m1p_ETH".RCV_VInp_05	BIN	2#0	
8		DB99.DBX 8.6	"PNOZ_m1p_ETH".RCV_VInp_06	BIN	2#0	
9		DB99.DBX 8.7	"PNOZ_m1p_ETH".RCV_VInp_07	BIN	2#0	
10		DB99.DBX 9.0	"PNOZ_m1p_ETH".RCV_VInp_08	BIN	2#1	
11		DB99.DBX 9.1	"PNOZ_m1p_ETH".RCV_VInp_09	BIN	2#0	
12		DB99.DBX 9.2	"PNOZ_m1p_ETH".RCV_VInp_10	BIN	2#0	
13		DB99.DBX 9.3	"PNOZ_m1p_ETH".RCV_VInp_11	BIN	2#0	
14		DB99.DBX 9.4	"PNOZ_m1p_ETH".RCV_VInp_12	BIN	2#0	
15		DB99.DBX 9.5	"PNOZ_m1p_ETH".RCV_VInp_13	BIN	2#0	
16		DB99.DBX 9.6	"PNOZ_m1p_ETH".RCV_VInp_14	BIN	2#0	
17		DB99.DBX 9.7	"PNOZ_m1p_ETH".RCV_VInp_15	BIN	2#0	
18		DB99.DBX 10.0	"PNOZ_m1p_ETH".RCV_VInp_16	BIN	2#1	

Fig. 35: Var Table – Virtual In/Outputs

4.5.2. Monitoring on PNOZmulti base unit

- Open the provided PNOZmulti-Project (“PNOZmulti_S7-PLC_ETH-TCP-IP_v2”) with Software PNOZmulti Configurator; the password for “Level 1” is number “1”.

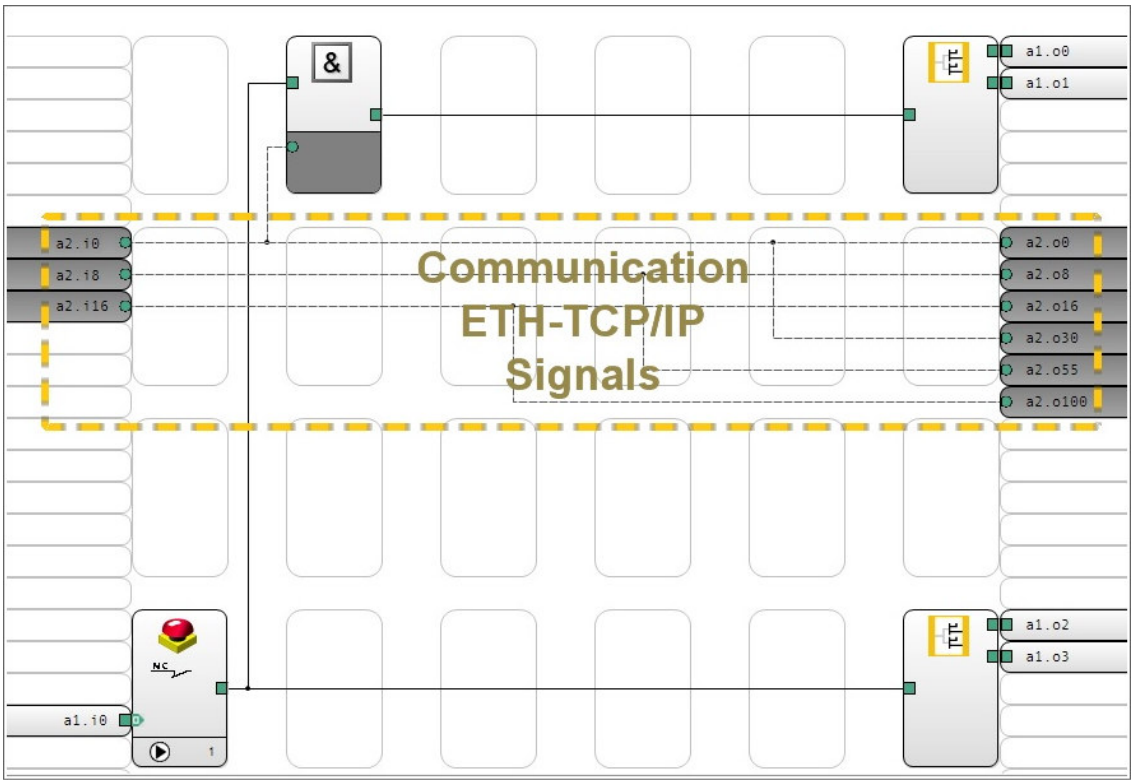


Fig. 36: PNOZmulti Configurator – Main Program “PNOZmulti_S7-PLC_ETH-TCP-IP_v2”

- Check the entered IP address (in example: 169.254.060.001), click at button “Online” and activate function “Dynamic Program Display”:

Fig. 37: PNOZmulti Configurator – Go “Online” and start “Dynamic Program Display”

If the lines of signals become green, than the Ethernet TCP/IP connection is online and the communication works.

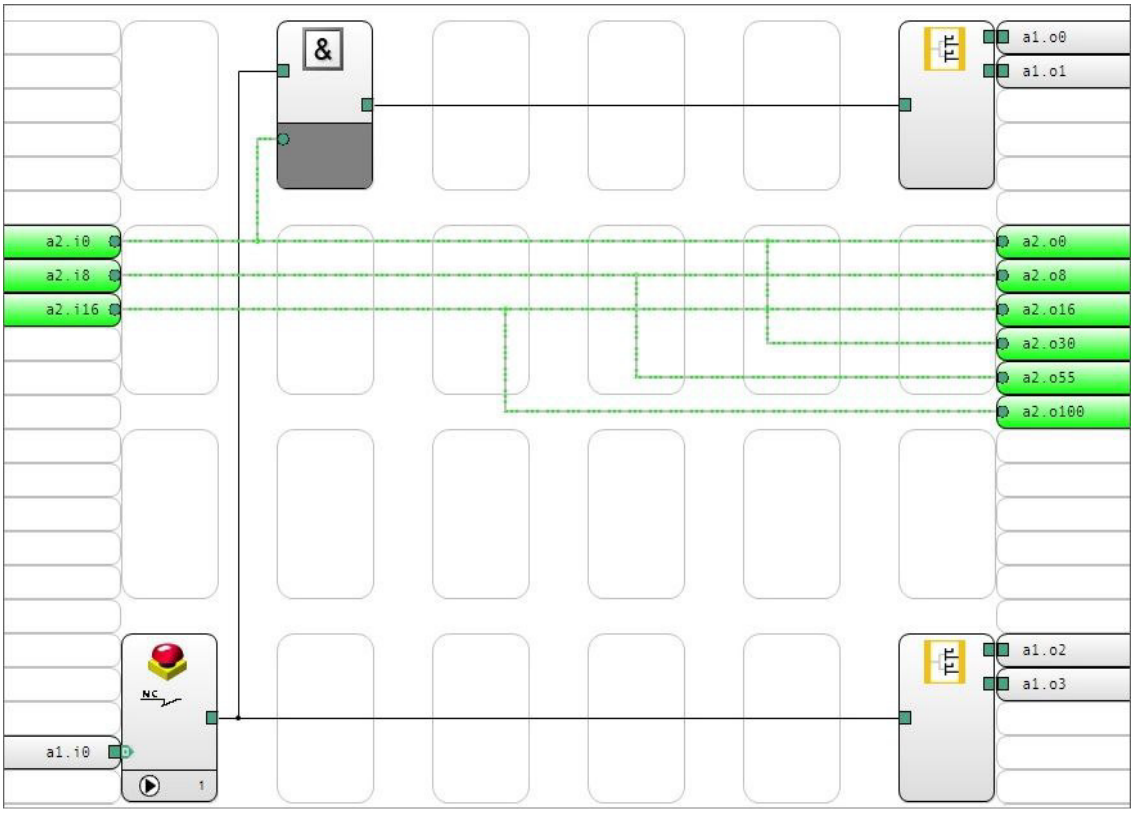


Fig. 38: PNOZmulti Configurator – Ethernet connection online and communication works

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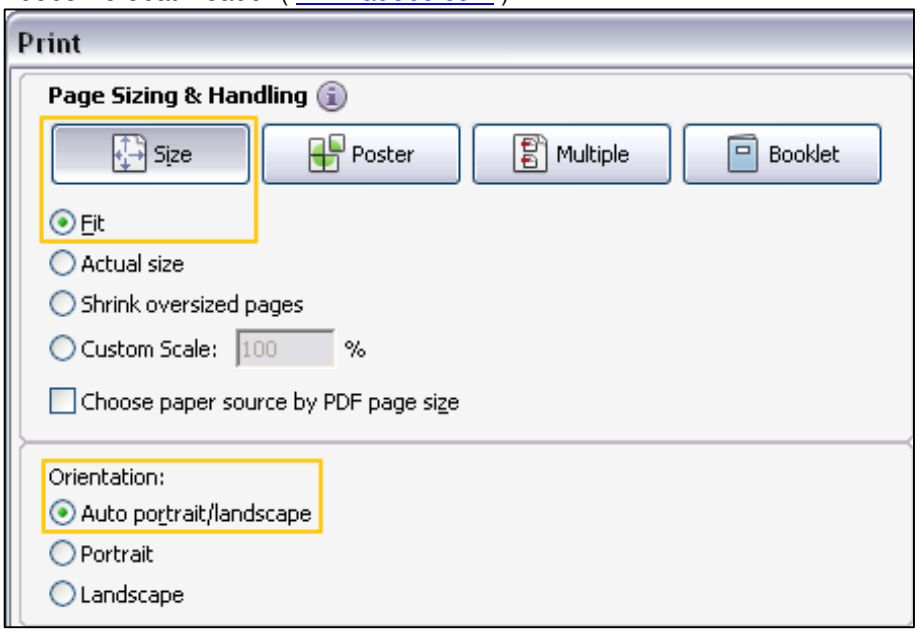
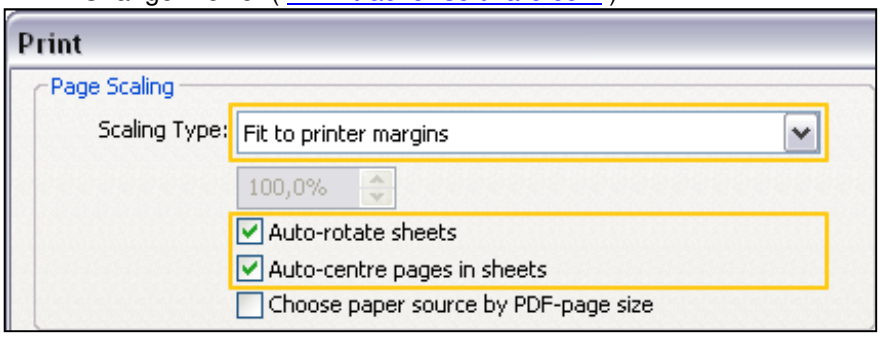
Recommended printer settings

Adobe Acrobat Reader (www.adobe.com)



PDF-XChange Viewer (www.tracker-software.com)



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Technical support
+49 711 3409-444
support@pilz.com

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In many countries we are represented by our subsidiaries and sales partners.

Please refer to our homepage for further details or contact our headquarters.

Pilz GmbH & Co. KG
Felix-Wankel-Straße 2
73760 Ostfildern, Germany
Telephone: +49 711 3409-0
Telefax: +49 711 3409-133
E-Mail: pilz.gmbh@pilz.de
Internet: www.pilz.com